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Ohio HAZUS-MH 2023 User Guide
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US Army Corps of Engineers®
Huntington District

Prerequisites

- AirGap computer with administrative permission on the default AirGap login
    - LRH ACE-IT is coordinating to have the HAZUS software install script modified so it will function properly
- ArcMap 10.8.2 installed with local license (Customize > ArcMap Options > General)
    - Relative path set as default
    - New layer set to not display
- HAZUS-MH 6.0 installed and OH state data downloaded
- uGet Download Manager installed (not needed with 1/3rd Arc Second DEMs)
- Pre-staged project counties layer, county watershed boundary files, county NSI files, and county Special Flood Hazard Area (SFHA) from the FEMA Flood Insurance Study (FIS), etc.
- Process videos have been previously provided

Process Standards

- DEM accuracy as decided by HAZUS PDT: 1/3rd Arc Second

  Multiple tests were done on the DEM but the 1/3rd arc second USGS DEM data has been selected. The 1/9th USGS DEM was tested but presented a smaller inundation limit in Franklin County and wasn't recommended. As a result of the tests new LiDAR based DEM data has been recommended recognizing the HAZUS watershed boundaries of all the counties. In addition, the DEM needed for HAZUS are in the IMG raster format and projected to match the USGS defaults. Tests showed that deviating from these standards cause problems in HAZUS. However, when Ashtabula County was done the USGS TNM portal only offers 1/3rd arc second files as TIF now, but it wasn't an issue during testing.

- Stream drainage area square miles: 4 SQ MI

  The stream drainage reference matches what was used in our previous HAZUS project but reviewing past projects show that a 2 SQ MI value could be used to better match exiting NFIP mapping in randomly selected areas. No testing was done to evaluate these differences for Ohio. This value affects a GIS process known as stream accumulation and determines the stream network headwater basin size. It is a professional opinion that 4 SQ MI is sufficient.

- USACE National Structure Inventory (NSI) fields

  The NSI public data was reviewed, and fields transformed for HAZUS. SHP files have been staged for each county and the NSI fields names modified to closely resemble CDMS fields. The SHP file fields are: State, County, GEOID, UID, Occupanc, BuildngR, ContentR, YearBult, AreaSqFt, NumStory, FloodFnd, FrstFlor, X, and Y. The NSI fields for Occupanc, FloodFnd, and NumStory were modified to match the field criteria in the CDMS. Occupanc was truncated to 5 characters, FloodFnd, was changed from initials to numerical values representing the matching CDMS value, and Number of Stories was tested for values > 8 and matching were replaced with 8. All the field names are 8 characters due to SHP file limitations but were changed to closely match CDMS matching tool field names for ease of use.

- Return periods used for damage results: 25r and 100r

The project return periods are based on Ohio's preference, but previous projects indicated a problem in HAZUS calculating return periods individually. The process was tested again and modified slightly based on the new version of HAZUS. Major stream networks in counties are run separately with a full suite of returns (10, 25, 50, 100, and 500) for each network. Then the 25r and 100r depth grids are used to build the final project results. This is due to how the software calculates the resulting depth grids.

- Data Management

  For our Corps HAZUS users a ProjectWise project has been created to share data amongst the team and a pre-staged work folder will be provided. The work folder has related GIS data and can be used the stage the DEM data needed for the county project. Other data will be available there such as the county FIS study, NSI SHP files, project counties, and HAZUS watershed boundaries for each county. More details related to data management will be in the process description below.

Summary

Counties will be assigned, and each person will be responsible for the HAZUS process, files generated, and results collected. There are three major phases described below:

1. Hydrology and Hydraulics (H&H)

   After creating a HAZUS project (ie: Franklin_HH) for your county and setting the Hazard > Flood Hazard Type to Riverine, the DEM files are downloaded using the pre-staged county watershed boundary file and creating a bounding box on the USGS The National Map portal. uGet is then used to download the files intersected in the portal tool and the text file list is then staged on your computer. After unzipping the IMG raster files, open HAZUS.

   Import the USGS DEM maps into the county project using the Hazard > User Data > DEM tab. Meta data is Meters and NAVD88. Import all the DEM files for your county and choose OK. After the RegionDEM is created choose the Hazard > Develop Stream Network command using the Stream drainage area square miles above (4 sq mi). Streams will be created from the DEM that are then used for the H&H analysis. The HAZUS user will need to evaluate the streams and group them be individual networks/basins. In the Franklin County example there were 4 networks located within the county boundary. Each network was used to create a scenario (ie: Net1, Net2, Net3, and Net4) and the Hazard > Riverine > Hydrology and Hazard > Riverine > Delineate Floodplain commands were run for each scenario. For a large number of stream networks like Ashtabula (9 networks), the Hydrology command can be run on all 9 scenarios before running the Delineate Floodplain on all nine. Failed reaches identified by the Hydrology command can be ignored. A full suite of return periods is used for in the Delineate Floodplain command (10, 25, 50, 100, and 500). Processing time is extensive depending on the size of the county and streams. The smallest network in Franklin took 14.5hrs with failed reaches being ignored. Ashtabula took 5 days to finish the entire process.

2. Combining County Network Depth Grids (GRIDS)

   A new project is created (ie: Franklin_UDF) similar as above and is used to combine the separate network (Net) scenario results for the 25r and 100r. The resulting depth grids from the HH project folder are used ie: Franklin_HH example: [C:\HazusData\Regions\Franklin_HH\NET1\Riverine\Depth]. The folder contains subfolders for each return period from the full suite of returns: rpd10, rpd25, rpd50, rpd100 and rpd500. They are in an ESRI Grid raster format and show as a standard list of files in a directory but display as raster when accessed using ArcMap. After setting the flood hazard type the Hazard > User Data > Depth Grid tab is selected. Add all the network rpd100 and rpd25 raster grids with their parameter set to feet

and a corresponding numeric return period value (ie: 100 or 25). Once imported create a scenario for each return period by selecting Hazard > Scenario > New. The New Scenario dialog appears, and the raster's are displayed on the screen. Select the entire county using the Add Select [+] button and the New Scenario dialog expands. Use the check mark button in the raster list table and select the corresponding raster files matching the scenario being created. Once the check marks are selected press the [Select] button on the bottom right of the form, save, and OK. Next choose Delineate Floodplain under Hazard > Riverine – verify and choose OK. Repeat for the remaining scenario and validate the differences between scenarios visually. The resulting CaseOutput.mdb > BoundaryPolyRP100 layer in the project HAZUS folder created for the 100r scenario will be used to extract the needed UDF structures from the corresponding staged county NSI SHP file.

3. User Defines Facilities (UDFs in CDMS)

   The USACE National Structure Inventory (NSI) data for Ohio will be used to update the OH HAZUS database using the Comprehensive Data Management System (CDMS) program. The pre-staged county NSI SHP files will be used for this phase. These files have all the structures for the county with unused field's deleted and used fields renamed to match the CDMS import process. The NSI fields are limited to 8 characters and do not match the 32bit fields in the CDMS but the field abbreviations match the CDMS words used. The structure location provides for a level 2 accuracy not found using the default level 1 Census data method.

   While the UDF project created above is open, insure the 100r scenario is loaded and verify the BoundaryPolgon path. Load the pre-staged NSI data for the county and Select by Location the NSI features using the BoundaryPolygon and a 0.1 mile buffer. Export the selected structures to a new SHP file (ie: Franklin_100r_NSI.shp) and visually verify the results. Export the resulting SHP file attributes to a spreadsheet using the same name and ArcMap command Table to Excel. IMPORTANT: the XLS file exported needs to be opened and saved again in Excel for it to work properly. Save the opened UDF spreadsheet (ie: Franklin_100r_NSI.xls) project and close. We are only adding 100r buffered structures that will be used for both 25r and 100r damage scenarios.

   Open the CDMS program and ensure the Ohio Statewide database is selected. Remove the history if present and choose Import into CDMS Repository from file. Unselect the Earthquake selection and browse for the 100r_NSI.XLS file exported above. Insure to change the extension to XLS in the Open File dialog in the CDMS tool to see the spreadsheet and select it. Choose the User Defines Facilities Category and Layer then Continue. Select No HAZUS ID then Continue. Match the fields as defined above. Once matched choose Continue and Yes at the prompt. The import occurs in two steps, after the initial import a message will say, "Data imported into CDMS repository". It also prompts to "…transfer data to Hazus-MH statewide datasets." Close the prompt and a UDF entry will show in the CDMS Repository windows. Select the "Transfer to Statewide Dataset" to initiate the transfer. If prompted to append or replace, choose Replace Data. Are you sure? Yes. Exit CDMS when finished. Choosing Replace data will keep the Ohio database small over all the users projects but this process would need redone if a project is done over.

4. Final Analysis

   The previous county projects will not have access to the UDF data so a new one will need to be created (ie: Franklin_FINAL). Once the county_FINAL project is created and opened check to insure the UDF data is accessible. Choose Inventory > User Defined Facilities and the User Defined Faculties dialog appears and should contain a table of structures. It now contains additional information that will be used for the damage cost estimates. There are additional buttons at the lower right if you choose to view them differently. If map is chosen the structures can also be viewed using GIS tools.

Two new scenarios will be created like the process above except the depth grids for the scenarios have already been merged. After selecting the Hazard > Flood Hazard Type > Riverine Only, select Hazard > User Data > Depth Grid and select the combined depth grids created above (ie: Franklin_UDF\25r\Riverine\Depth\rpd25). Set corresponding parameters and add the next depth grid > Choose OK. Two new scenarios are created as above and the Riverine > Delineate Floodplain executed. Visually evaluate both results. After these steps are performed an analysis is executed for each scenario as follows.

With the scenario selected and confirmed, select the Analysis > Run command, choose the Select All button and OK to run. When complete, choose the next scenario and run. Once both scenarios are completed, all the commands under the Results menu item can be viewed and reports exported.

5. Exporting UDF files, Reports and Archiving HAZUS Project

The final project file (ie: Franklin_FINAL) contains the depth grids, the UDF structures, and the analysis that was run for the final UDF results for the r25 and r100 scenarios. There will be four files exported 2 UDF files and 2 report summary files for each scenario. The active scenario can be confirmed by at the upper left corner of the GIS project window – the project file name is shown next to the globe. After the project file name in parentheses is the active scenario. Scenarios are changed using the Hazard > Scenario > Open command and after a new scenario is selected the Results > View Current Scenario Results By > Available Results is used to populate all the result selections available.

Once the scenario results are populated the selections that will be exported are the User Defined Facilities (UDF) spreadsheet and the Flood Global Risk Report. The file default names will be prefixed with the return period and the county (ie UDF: 100r_Franklin_absv_FRUserDefinedFlty.txt and ie Summary Report: 100r_Franklin_hzflcr_gsr.pdf) and saved to the [OH-HZ\OUT\Franklin] staging directory. Using these naming conventions will group the return period files together.

To export the UDF file select the scenario under Hazard > Scenario > Open. Confirm the scenario in the upper left project window. Select Results > View Current Scenario Results by > View Results by > confirm scenario and choose OK. All reports are now available. Choose Results > User Defined Facilities > Right mouse click in window > Export. Navigate to OUT staging directory and prefix file as described above. Close UDF window and go to Results > Summary Reports > Other tab > select Flood Global Risk Report and choose View – this will generate the PSF file and takes some time. When the report is finished it will pop-up in a new window. Choose the upper left menu icon to save the file > choose OK > All pages > OK. Navigate to OUT staging directory and prefix file as described above. Repeat for the 25r scenario and save project and close.

The last export to be accomplished for completing a county project is to use the initial Hazus Startup menu to Export/Backup a region. After launching HAZUS choose Export/Backup a region > choose the county project final (ie: Franklin_FINAL) > select Browse > navigate to your desktop. The export will not write to your external hard drive. Once complete close and exit HAZUS and move the file to your OUT staging directory and prefix the file with [ZIP_] ie: ZIP_Franklin_FINAL.hpr > Choose Export/Backup. This saves the entire project to an HPR file that is in a ZIP compressed file format. The final deliverable to Ohio EMA will be all these county files along with a summary file that will be created using Excel using the UDF results for each county.

6. OH HAZUS Project Directories

A project directory structure has been created to facilitate the HAZUS project creation for the 25 counties.

Folder1	Folder2	Folder3	Note
OH-HZ			Contains all folder's (level 2 and 3)
	DEM		Staging directory for DEM data
	OUT		Staging directory for all project files
	REF		Contains all folder's (level 3)
		Coastal	Working directory for counties on Lake Erie
		NSI	OH NSI SHP files for project counties
		SFHA	OH SFHA SHP files for project counties
		SHP	Project SHP files and Project Counties
		VIDS	Process videos
		WB	Project ReqDEMPolygon for Project Counties
	wrk		scratch folder for data file management

This folder structure with all reference files will be ZIPPED and deliver to our HAZUS users. This is intended to organize the project and make the process of creating the projects and finalizing the data easier. The above table is self-explanatory and describes how they are used and what they are for. File naming conventions have been discussed previously and are intended to be collected in these same folders for a final product delivery to OH OEM.

The OH-HZ | REF | folders contain files needed in the project creation process specially the NSI, SFHA, and WB folders. The SHP folder contains project level reference files and currently contains a Project_Counties.shp file layer. The remaining directories contain SHP files for each county supporting the process described above. The NSI file is used for the UDF process, the SFHA file is referenced when breaking out and analyzing the stream networks, and the WB file is the HAZUS created project watershed file used when determining the DEM files needed for HAZUS. This document will also be placed in the main OH-HZ folder for reference.

The OH-HZ | OUT | folder is intended to have the subdirectory's of assigned counties of our technical POC's. As per our instructions these folders should contain the following files as an example:

1. 100r_Franklin_absv_FRUserDefinedFlty.txt
2. 100r_Franklin_absv_FRUserDefinedFlty.txt.log
3. 100r_franklin_hzflcr_gsr.pdf
4. 25r_Franklin_absv_FRUserDefinedFlty.txt
5. 25r_Franklin_absv_FRUserDefinedFlty.txt.log
6. 25r_franklin_hzflcr_gsr.pdf
7. Franklin_100r_NSI.cpg
8. Franklin_100r_NSI.dbf
9. Franklin_100r_NSI.prj
10. Franklin_100r_NSI.sbn
11. Franklin_100r_NSI.sbx
12. Franklin_100r_NSI.shp
13. Franklin_100r_NSI.shx

14. Franklin_100r_NSI.xls
15. Franklin_FINAL.hpr

Based on this Franklin County example with no errors there are 15 files that represent the UDF output files for both the 25r and 100r, the Flood Global Risk Report for the 25r and 100r (PDF), the backup of the county HAZUS Final project (HPR), the county 100r NSI SHP layer group of files, and the associated 100r NSI XLS spreadsheet file. Not shown in this example is ERROR.txt mentioned in the error paragraph below.

To share and archive the results the entire OUT directory for each county will be zipped and uploaded to the designated project location in ProjectWise as coordinated by the internal Corps team. Maintain all the working files on your hard drive/HAZUS computer and await further instructions on how they will be managed. Right-mouse click on the county folder in the [OUT] directory and choose 7-Zip > add to "Franklin.7z". The 7z format is a Corps standard and provides for better compression on the archive. Append the county name to look like the following: OH_HAZUS_FINAL_Franklin.7z.

7. Coastal Counties

Coastal counties refer to the northern most counties that adjoin Lake Erie: Ashtabula, Cuyahoga, Lake, and Lorain. In the HAZUS-MH program there is a reference to riverine and coastal and up to this point riverine has been selected and used for all the processing. However, HAZUS-MH lets you create coastal transect layers that create depth grids and return periods. Upon reviewing the HAZUS-MH coastal process an alternate process was used. An alternant method external to HAZUS-MH was used that included downloading the Flood Insurance Study FIS report and NFHL Data for all jurisdictions in the county. The FIS report contains a table entitled, Coastal Transect Parameters (CTP) that includes return periods for all the transects (cross sections for a coast). In addition, the NFHL Data includes a layer entitled S_CST_TSCT_LN.shp. This SHP file contains all the transects but only the 1% annual value in the field named "SWEL". This allowed for an alternate GIS method to be used and to create depth grids outside of HAZUS-MH as a regular GIS project.

All the reports and data were downloaded for each coastal county and reviewed. The FIS reports for each county contained the CTP table that included annual chance values of 10%, 4%, 2%, 1%, and 0.2%. The 4% annual chance column was present but did not contain data. These annual percent values correspond to 10-year, 25-year, 50-year, 100-year, and 500-year like a full suite of returns in HAZUS-MH. However, the 25-year needed for our project was not available. To overcome this missing 4%/25-year data a mathematical interpolation formula was used to solve for the 25-year. The formula as referenced was $y = y1 + (x - x1) \times (y2 - y1) / (x2 - x1)$. In this formula, x1, x2, y1 and y2 are known values but by adding the 25-year return period for x, the resulting missing value can be calculated based on the values of the 10-year and 50-year. This formula as written in the spreadsheet was = E2 + (0.04 - 0.1) * (G2 - E2) / (0.02 - 0.1) where the return periods were in written in decimals and the Excel columns represented the adjoining columns in the 4%/25-year/0.04.

Each CTP table was exported as PDF pages and converted to an Excel Spreadsheet. The converted result was cleaned and formatted to resemble a database table. The 25-year column and formula were added, and the results calculated. All formulas were converted to number results and a TXT file was exported. ArcMap can import TXT files but not XLSX files. After preparing the Excel spreadsheet GIS was used to create the depth grid. A GIS project file was created and the S_CST_TSCT_LN.shp file was added from the NFHL Data. Two modifications were added: the lines were extended out from each right and left direction and two additional transects were added to each end of the series. The transects were extended to take in more area focusing on channel inlets and to ensure the depth grid extended out beyond the county boundary. The added transects were the same value as the first and last transects.

Once the transect layer is edited and modified the spreadsheet TXT file is attached to the GIS project file. The Excel TXT table is joined to the S_CST_TSCT_LN layer using the TRAN_NO field. Once joined all the fields from the layer and the TXT table file can be seen in the layers attribute table. This layer is then exported to a new SHP file and the fields deleted that are not needed. The fields retained contain the Transect number, and the elevation values for the 1%/100-year and the 4%/25-year. This layer is then used to create a TIN file for each return period that is then exported as Raster files. When creating TIN surfaces from the updated Transects make sure the same projection is used as for the counties DEM file as well as the pixel size.

Once the water surface grids are created the GIS raster calculator is used to create two files. One of the raster files is a depth grid that is created by subtracting the DEM from the water surface. The other is a mask layer created by exporting all the values greater than 0 from the raw depth grid. The mask raster file is then further exported as a polygon file and creates boundaries that are 0 and 1 based on raster math. The SHP file exported has a gridcode that contains 0 and 1. The SHP files is then edited to delete all the records with a value of 0. This leaves a polygon file that is considered the inundation limits for the depth grid and is also used as a mask. At this point the Extract by Mask command is used to create the final depth file for both the 25r and 100r.

These depth grids were then used and imported into the Ashtabula County project just as described in paragraph 2 - Combining County Network Depth Grids (GRIDS). When this is done the selection for Riverine is still used for the active project and after going through the remaining steps the results will include 25r and 100r damages from the Lake Erie as based on the FIS study results.

8. HAZUS Errors

After starting into the production phase, a number of errors presented themselves that were not identified in pre-production. There are currently two types of errors: one that occurs when doing the User Defined Facilities Analysis and another when importing the NSI spreadsheet into the CDMS. The UDF analysis error results vary but the initial error indicates that the Occupancy, NumStories, and Foundation Type is NULL. Research indicates that they are not null and affects the results in two way. The first is that it uses default values for these fields and the second is that there is no damages calculated. These errors do not keep results from being calculated.

The second known error occurs when importing the UDF spreadsheet using the CDMS tool. The error is: “Object reference not set to an instance of an object” and the structures are not imported. Currently there are no fixes identified for these errors but when these errors occur, they need to be document in the counties OUT folder.

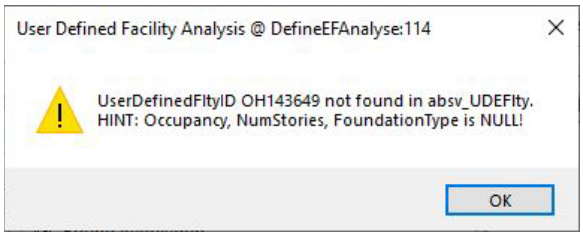
Any error that occurs should be documented by creating a text file in the counties OUT folder called ERROR.txt. This can be created with windows Notepad program a d it should begin with your name and date of the error. Write a short description of the error and if more than one error occurs, add an additional date to the file and add the additional description. If you are able to continue running the process, then do so. We may find solutions to these error at a later date and can return to fix them.

If you are not able to continue, then it is vital to archive your current progress using the Backup/Export Region command. These errors are occurring after the H&H phase and it’s important that your backup includes the r25 and 100r scenarios if they were completed. The idea is to continue processing the counties and if a solution is found we may be able to return to these counties and fix/finish the process.

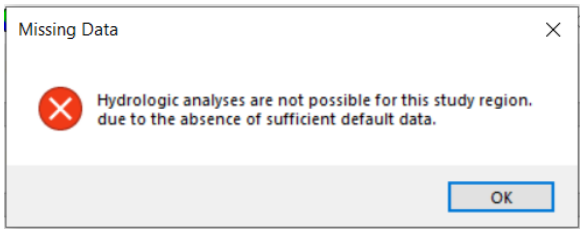
9. HAZUS Error Post Processing Summary

During the process of development there were minor errors that were noticed that were considered native to the HAZUS-MH process. These issues were not researched or documented in detail but included: stream network reaches not being included after selection as well as user defined facilities (UDF) being identified as not found during the analysis process. Problem Reaches as identified by HAZUS were accepted and the remainder of the process continued. These are described as an issue between the DEM used and the available hydrology table references used in the process. For example, on Hamilton County, an entire river reach was returned as problem reaches but when spilt into smaller individual reaches, the delineation of the flood plain occurred on more of the SFHA reaches identified. The UDF Not Found issue was not researched in the HAZUS documentation, but an effort was made to determine why this was occurring since our process included uploading custom UDF data using the CDMS tool.

Reviewing the UDF error included looking at the specific records and finding that for unknown reasons HAZUS would replace fields with default data. The fields were Occupancy, NumStories, and FoundationType, and were identified as not found.



Comparing the results showed that these records had default data attributed to them automatically in the process. Consequently, these errors were also accepted as part of the typical HAZUS process like the problem reaches. In addition, one of our HAZUS users experienced the error: “Hydrologic analyses are not possible for this study region due to the absence of sufficient default data.” Her fix was to reinstall HAZUS.



In addition to these accepted errors, there were two additional errors that required more research and changes in the process. These errors were: different projections in the stream networks in Greene County and a UDF import error in Cuyahoga and Lorain counties. The different projections manifested in the H&H phase of Green County, but the error occurred in the UDF phase when the depth grids are merged. The fix in this situation was to merge the raster data outside of the HAZUS program using ArcMap. These merged rasters were then imported into the UDF phase of the project as described in the process.

The UDF import error was the most problematic and was saved until all the rest of the counties had been completed. The fix for Cuyahoga County was different than Lorain County but after all the counties were completed, time was spent on identifying the specific error and coordinating with the HAZUS Help Desk. An analysis was done on the Cuyahoga NSI spreadsheet data that included deleting records and then importing them into CDMS. This was done until a single record was left that caused an import error. Coordination with the HAZUS help desk included receiving a new CDMS version but this was not used. For Cuyahoga County the fix was to delete this specific record.

The approach to fix the Lorain County CDMS import error was different. After finding a one record problem in Cuyahoga using a long process of elimination, a new approach was used. In our documented process, a 0.1-mile buffer is used with the 100r return period to extract the NSI data set used for importing the structures. This process was modified for Lorain to use the 100r boundary without the buffer. Buffers are good practice in GIS processes but with the quality hydrologic process of defining networks and using a full suite of return periods based on the same hydraulic data, it was theorized that issues would be miner and isolated. The unbuffered NSI data imported without error.

The documented process is intended to provide a sort of quality assurance in the results and one of the checks includes compiling the results in a Microsoft access database to review the damage data. This is not a significant quality check, but only one issue was observed, and it didn't include Cuyahoga or Lorain. In Greene County there is a single COM6 structure identified in the 25r data that is not present in the 100r. It's present in the 25r data but has zero values on the damage results. It does not show any dollar values for damages.

Final Comments

To summarize, the HAZUS process was broken into three major phases for each county with a HAZUS project created for each phase: 1) hydrology and hydraulics running full suite of return periods on SFHA identified stream networks (HH), 2) merging stream networks into 25r and 100r scenarios and extracting NSI structures using a 0.1-mile buffer from the 100r into CDMS as user defined facilities (UDF), and 3) creating scenarios and calculating the damages for the two scenarios. In some cases, additional projects were created to fix issues in the H&H phase. File management conventions were stablished for reference in the process as well as providing a standard set of output files. The process was considered the basis for quality control with the final review consisting of a resulting data review. The data review consisted of two methods: compiling all the county results into a single Microsoft Access database that allowed for viewing and comparing the damage results as well as revieing the Final HAZUS project to ensure the results contained the appropriate structures in the scenario's inundation limits. In addition, the county CDMS UDF inventory was imported into the database and used to validate the results using the HAZUS unique ID for each structure (tbl_CDMS_UDF_Inventory). This was used to compare both scenario results (25r and 100r) using the building value and content value fields. The query results show that each scenario had structures in the HAZUS results that were not in the CDMS inventory. The 25r comparison (q_CDMS_HAZUS_Compare_25r) contained 12 across 8 counties and the 100r comparison had 14 across 8 counties. This reason for this inconsistency is unknown.